

ASSIGNMENT-4

22BCON1068
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Date: 14/11/24 31

Subject: Computer Graphics

• Section - A (2x5=10)

1. What is 3D Translation?

Ans 3D translation is shifting an object along the x, y and z axes by specified distances (T_x, T_y, T_z). The transformation moves every point of the object without changing its orientation or size.

Matrix Representation:-

$$\begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. What are geometric transformations?

Ans Geometric transformations are operations that modify the position, size, orientation or shape of objects in a coordinate space.

3. What is 3D Scaling?

Ans 3D scaling changes the size of an object along the x, y, and z axes by specified factors (S_x, S_y, S_z), expanding or shrinking it proportionally.

Matrix representation:-

$$\begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. Give Transformation matrices of all type of shearing in 3D.

Ans In 3D shearing, objects are skewed along one axis while keeping the others fixed. There are six types of 3D shearing transformations:-

(i) Shear along X-axis using Y and Z:

$$\begin{bmatrix} 1 & S_{xy} & S_{xz} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(ii) Shear along Y-axis using X and Z:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ S_{yx} & 1 & S_{yz} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(iii) Shear along Z-axis using X and Y: -

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ S_{zy} & S_{yx} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Q. What are Homogeneous coordinates?

Ans Homogeneous coordinates are an extension of regular Cartesian coordinates used in computer graphics, adding an extra dimension (w) to represent points and transformations more easily. A point (x, y, z) becomes (x, y, z, w). This allows for translations, projections, and other transformations to be expressed as matrix multiplications for 2D, a

Section - B

Q. Transform the given point P(10, 10, 10) into 3D space with translation factor T(10, 20, 5) and scaling factor S(3, 2, 5).

Ans given point

P(10, 10, 10) in 3D space with translation factor T(10, 20, 5) and scaling factor S(3, 2, 5).

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Firstly we calculate scaling Transformation:-

(i) The scaling matrix for scaling factors $s_x = 3$, $s_y = 2$ and $s_z = 5$ is:

$$\begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

To scale the point $P(10, 10, 10)$:-

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \times 10 \\ 2 \times 10 \\ 5 \times 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \\ 50 \\ 1 \end{bmatrix}$$

So After scaling we get the point $P'(30, 20, 50)$.

(ii) Translation transformation:

transl. matrix for transl. factors $T_x = 10$, $T_y = 20$, $T_z = 5$ is

$$\begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

To translate the scaled point $P'(30, 20, 50)$:

$$\begin{bmatrix} x'' \\ y'' \\ z'' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 30 \\ 20 \\ 50 \\ 1 \end{bmatrix} = \begin{bmatrix} 30+10 \\ 20+20 \\ 50+5 \\ 1 \end{bmatrix} = \begin{bmatrix} 40 \\ 40 \\ 55 \\ 1 \end{bmatrix}$$

So $P''(40, 40, 55)$

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(ii)

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(ii) Explain 3D reflection relative to all 3 planes.

Ans 3D reflection is a transformation that flips an object across a specific plane, producing a mirror image. In 3D space, reflections can be done relative to the xy plane, yz plane or xz plane.

(i) Reflection with respect to the xy plane :-

- Effect: - The z -coordinate of the point changes sign, while the x and y coordinates remain the same.
- Matrix representation: -

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Example: point $(x, y, z) \rightarrow (x, y, -z)$.

(ii) Reflection with respect to the yz plane :-

- Effect: - The x -coordinate of the point changes sign, while the y and z coordinates remain the same.
- Matrix representation: -

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Example: point $(x, y, z) \rightarrow (-x, y, z)$.

(iii) Reflection with respect to xz plane.

- Effect: - The y -coordinate of the point changes sign, while the x and z coordinates remain the same.

- Matrix representation: -

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$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example :- Point $(x, y, z) \rightarrow (x, -y, z)$.

Q9. Explain homogeneous coordinates for translation and scaling.

Ans Homogeneous coordinates are a system in which an extra dimension (w) is added to the Cartesian coordinate to represent points. This allows w to express translations and other transformations using matrix multiplication, which is not possible in standard Cartesian coordinates.

In 3D space, a point (x, y, z) is represented as $(x, y, z, 1)$ in homogeneous coordinates.

• Translation in Homogeneous Coordinates :-

In Cartesian coordinates, translation involves adding values to the coordinates, which cannot be directly achieved using matrix multiplication. However in homogeneous coordinates, we can express translation with a 4×4 matrix.

Translation Matrix :-

$$\begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Here, T_x , T_y , and T_z are the translation factors along the x , y and z axes, respectively.

• Applying Translation :- For a point $P(x, y, z, 1)$:

$$\begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x + T_x \\ y + T_y \\ z + T_z \\ 1 \end{bmatrix}$$

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- Result: - The point is Translated to $(x+T_x, y+T_y, z+T_z)$.

- Scaling in Homogeneous coordinates:-

Scaling changes the size of an object along the x , y and z axis. In homogeneous coordinates, we can express scaling using another 4×4 matrix.

Scaling matrix:-

$$\begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Applying scaling: For a point $P(x, y, z, 1)$:-

$$\begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} S_x \cdot x \\ S_y \cdot y \\ S_z \cdot z \\ 1 \end{bmatrix}$$

Result:- The point is scaled to

$$(S_x x, S_y y, S_z z)$$

- Sec-C (3x11=33)

1. A 3D triangle $A(3, 4, 2)$ $B(6, 4, 2)$ $C(5, 6, 3)$ is reflected on yz plane. Find the new coordinates and rotate it by 90 degree around z -axis.

Ans Step 1: Reflecting on the yz Plane

Reflecting a point on the yz plane changes the sign of its x -coordinate, while leaving the y and z coordinates unchanged.

Reflection matrix:- (yz Plane):-

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Applying the Reflection:-

- for $A(3,4,1)$, after reflection - $A'(-3,4,1)$
- for $B(6,4,2)$, after reflection - $B'(-6,4,2)$
- for $C(5,6,3)$, after reflection - $C'(-5,6,3)$

STEP 2: Rotating by 90° around the z-axis.Rotating a point by 90° counter clockwise around the z-axis changes the coordinates as follows:-

$$(x, y, z) \rightarrow (-y, x, z)$$

Rotation matrix (90° around z-axis):-

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Applying the Rotation -

- for $A'(-3,4,1) \rightarrow A''(-4,-3,1)$
- for $B'(-6,4,2) \rightarrow B''(-4,-6,2)$
- for $C'(-5,6,3) \rightarrow C''(-6,-5,3)$

Ans

Q2. Given a 3D triangle with points $(0,0,0)$, $(1,1,2)$ and $(1,1,3)$.

APPLY shear parameter 2 on x axis, 2 on y axis and 3 on z axis and find out the new coordinates of the object.

Ans

Given:-

Original vertices:-

$$A(0,0,0)$$

$$B(1,1,2)$$

$$C(1,1,3)$$

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Shear Parameters:-

Shear along x-axis: $S_{xy} = 2$, $S_{xz} = 3$ Shear along y-axis: $S_{yx} = 2$

Shearing matrix:-

$$\begin{bmatrix} 1 & S_{xy} & S_{xz} & 0 \\ S_{yx} & 1 & 0 & 0 \\ S_{zx} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

So matrix:-

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Applying the Shear Transformation:-

(i) Point $A(0,0,0)$:-

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(0) + 2(0) + 3(0) + 0 \\ 2(0) + 1(0) + 0(0) + 0 \\ 0(0) + 0(0) + 1(0) + 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

New Coordinates: $A'(0,0,0)$ (ii) Transforming Point $B(1,1,2)$:-

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(1) + 3(2) + 0 \\ 2(1) + 1(1) + 0(2) + 0 \\ 0(1) + 0(1) + 1(2) + 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+2+6 \\ 2+1+0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

New Coordinates:- $B'(9,3,2)$ (iii) Transforming point $C(1,1,3)$:-

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(1) + 3(3) + 0 \\ 2(1) + 1(1) + 0(3) + 0 \\ 0(1) + 0(1) + 1(3) + 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+2+9 \\ 2+1+0 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 12 \\ 3 \\ 3 \\ 1 \end{bmatrix}$$

New Coordinates: $C'(12,3,3)$ So Axis $A'(0,0,0)$, $B'(9,3,2)$, $C'(12,3,3)$.

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Q3. Given a line between $(10, 5, 6)$ and $(20, -5, 10)$. Perform the following transformations in sequence:

- Translate it by $T(5, 2, -10)$.
- Rotate it by 90° in clockwise dir. along y axis.
- Scale it to 2 times its size.
- Retract it along the x-z plane.

Ans. Given line endpoints: $(10, 5, 6)$ and $(20, -5, 10)$.

(a) Translate by $T(5, 2, -10)$.

translation matrix for $T(5, 2, -10)$ is

$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculation - $(x', y', z') = (x + T_x, y + T_y, z + T_z)$

• for $(10, 5, 6) \rightarrow (10 + 5, 5 + 2, 6 + (-10)) = (15, 7, -4)$

• for $(20, -5, 10) \rightarrow (20 + 5, -5 + 2, 10 + (-10)) = (25, -3, 0)$

(b) Rotate it by 90° in clockwise around y-axis.

for a 90° clockwise rotation around the y-axis

the transformation matrix is

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculation: - $(x', y', z') = (z, y, -x)$

for

for $(15, 7, -4) \rightarrow (-4, 7, -15)$

for $(25, -3, 0) \rightarrow (0, -3, -25)$

(c) Scale it to 2 times its size.

The scaling matrix for scaling factor of

$S_x = 2, S_y = 2, S_z = 2$ is

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$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Scaling Calculation :- $(x', y', z') = (2x, 2y, 2z)$ for $(-4, 7, -15) : (-8, 14, -30)$ for $(0, -3, -25) : (0, -6, -50)$ (d). Reflect along the $x-z$ Plane.The reflection matrix for the $x-z$ Plane (which inverts y -coordinate) is :-

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Reflection Calculation :- $(x', y', z') = (x, -y, z)$

So Calculation

for $(-8, 14, -30) = (-8, -14, -30)$ for $(0, -6, -50) = (0, 6, -50)$

So After applying all the transformations (translation, rotation, scaling and reflect) the new coordinates of the line segment are :-

Point 1 :- $(-8, -14, -30)$ Point 2 :- $(0, 6, -50)$.

x

x

x

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